Storage Interfaces and Disk Standards

Introduction

Modern computer systems use different **storage interfaces** to connect disks and solid-state drives to servers and client systems. These interfaces affect:

- **Data Transfer Speed**
- **Latency** (delay in accessing data)
- **Reliability and Scalability**

Disk Interface Standards Families

1. SATA (Serial ATA)

- Designed for consumer devices (PCs, laptops).
- Uses a simple, low-cost connection.
- **SATA 3** supports speeds of up to **6 Gb/s**.
- Primarily used in HDDs and entry-level SSDs.

2. SAS (Serial Attached SCSI)

- Enterprise-grade interface for servers and data centers.
- Offers high reliability, advanced error checking.
- **SAS v3** supports speeds of up to **12 Gb/s**.
- Can connect multiple devices in a daisy-chain.

3. NVMe (Non-Volatile Memory Express)

- Optimized for SSDs, bypassing legacy disk protocols.
- Works over the **PCIe (Peripheral Component Interconnect Express)** bus.
- Offers **very low latency** and extremely high performance.
- Supports data transfer rates of up to **24 Gb/s**.
- Ideal for cloud, databases, and high-performance applications.

Comparison Table: SATA vs SAS vs NVMe

	SATA	**SAS**	**NVMe**
Max Speed	6 Gb/s (SATA 3)	12 Gb/s (SAS v3)	24 Gb/s (PCIe 4.0)
Latency	Moderate	Lower than SATA	Very low (microseconds)
Typical Use	Consumer PCs, laptops	Enterprise servers, RAID	Cloud, HPC, databases
Devices	HDDs, SSDs	HDDs, SSDs	Only SSDs
Cost	Low	Medium	High

SSD Performance Metrics

IOPS (Input/Output Operations Per Second)

- **Typical 4 KB Reads:** ≈ 10,000 IOPS (reads per second).
- **Typical 4 KB Writes:** ≈ 40,000 IOPS.
- SSDs support parallelism, significantly improving performance.

Parallel Performance (Queue Depth = 32)

- **Reads:**
  - $\approx$ 100,000 IOPS with QD-32 on SATA.
  - $\approx$ 350,000 IOPS with QD-32 on NVMe PCIe.
- **Writes:**
  - $\approx$ 100,000 IOPS with QD-32.
  - Higher values possible on advanced NVMe SSDs.

Sequential Data Transfer Rate

- **SATA3 SSDs:** Up to 400 MB/s.
- **NVMe PCIe SSDs:** 2–3 GB/s (significantly faster).

Hybrid Disks

- Combine a small flash cache with a larger magnetic disk.
- Flash cache is used for frequently accessed data, improving responsiveness.
- Provides a balance between low cost (HDD) and improved performance (SSD).

Direct vs. Networked Storage

Direct-Attached Storage (DAS)

- Disk is connected **directly** to a computer system.
- Examples: Internal HDDs, SSDs, external USB drives.
- Fast and simple, but not easily shared between multiple servers.

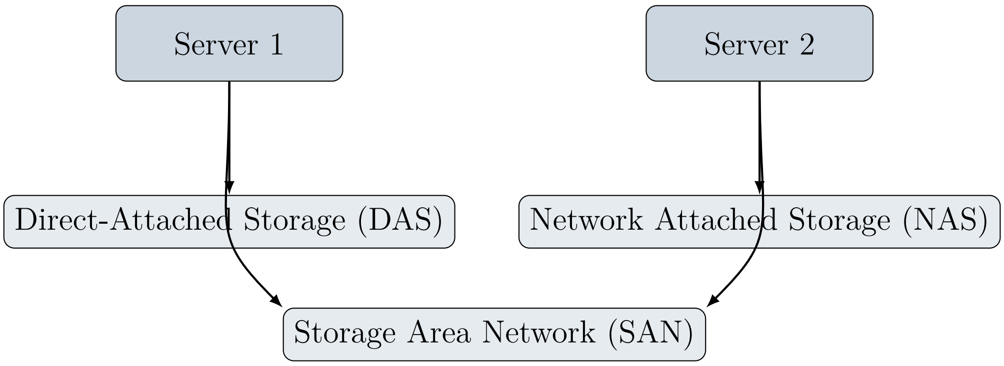
Storage Area Network (SAN)

- **Block-level storage** shared across servers.
- Uses a **high-speed network** (often Fibre Channel or iSCSI).
- Provides centralized management and scalability.
- Used in large enterprises and data centers.

Network Attached Storage (NAS)

- Provides a **file system interface** over the network.
- Uses protocols such as **NFS (Network File System)** or **SMB (Server Message Block)**.
- Easier to set up than SAN, widely used in offices and homes.

Storage Architecture Diagram



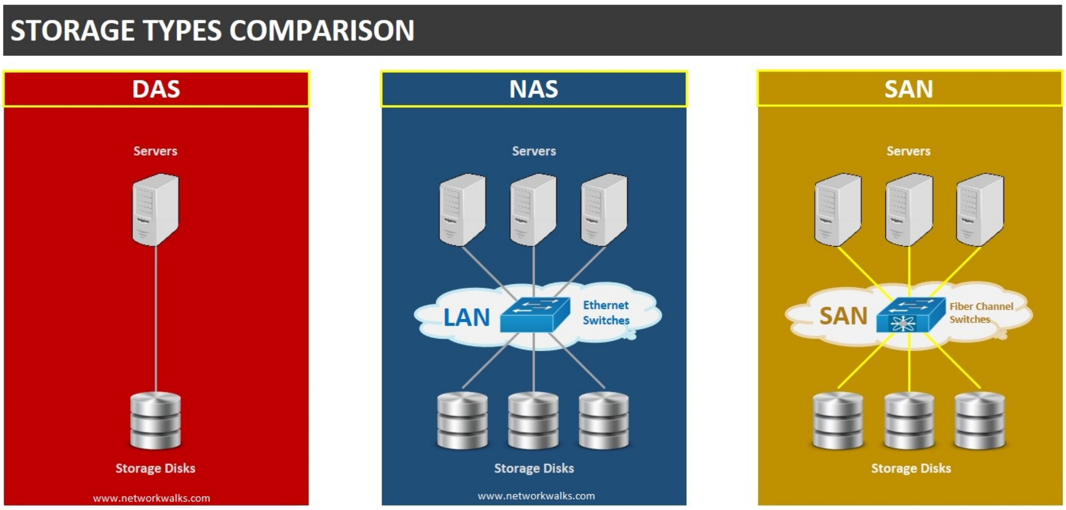


Figure 1: Comparison of Direct Attached Storage (DAS), Network Attached Storage (NAS), and Storage Area Network (SAN).

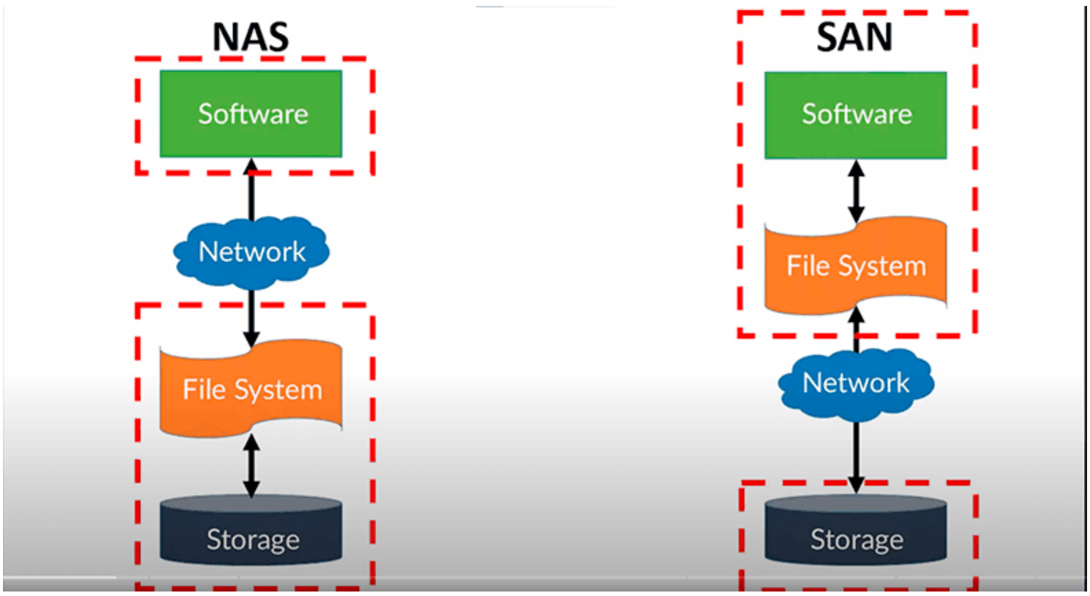


Figure 2: Network Attached Storage (NAS), and Storage Area Network (SAN).